



SECJ2203: Software Engineering

System Documentation (SD)

Project KADA eServe

Version 1

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Faculty of Computing

Prepared by: TechAway

Revision Page

a. Overview

KADA eServe is a local database-driven document management platform developed for Lembaga Kemajuan Pertanian Kemubu(KADA) to replace the manual processes. It manages key processes such as member registration, loan applications, and report generation, ensuring secure and efficient data handling. This system is operated by clerks, who update application statuses based on decisions made by Board of Directors (BOD) conveyed during meetings. This

b. Target Audience

This document is intended for following audiences:

- 1) System Developers: To understand the detailed requirements and functions for building the project
- 2) Project Manager: To ensure project deliverables align with the documented requirement
- 3) System Tester: To execute test cases for ensuring the system meets all requirement
- 4) KADA Management and Board of Directors: To understand system's capabilities and approve its features
- 5) Clerks: To gain clear understanding of the system's features and benefits
- 6) KADA Users and Members: To familiarize themselves with system processes and functions

c. Project Team Members

Member Name	Role	Task	Status
Poh Lok Yee	Leader	- Specify design constraints - Document launch phase - Create user story - Draw sequence and activity diagram	Complete
Brendan Chia Yan Fei	System analyst	- Create user story - Draw sequence and activity diagram	Complete
Gui Kah Sin	System testing	- Create user story - Draw sequence and activity diagram	Complete
Sabrina Heng Wei Qi	System designer	- Create user story - Draw sequence and activity diagram - Define performance and other requirement	Complete
Tan Zhi Ming	System developer	- Create user story - Draw sequence and activity diagram - Define performance and other requirement	Complete

d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Poh Lok Yee Brendan Chia Yan Fei Gui Kah Sin Sabrina Heng Wei Qi Tan Zhi Ming	Completed SRS	31/12/2024

Table of Contents

1	Introduction		1
	1.1	Purpose	1
	1.2	Scope	?
	1.3	Definitions, Acronyms and Abbreviations	?
	1.4	References	?
	1.5	Overview	?
2	Specific Requirements		?
	2.1	Persona	?
		2.1.1 KADA User 2.1.2 KADA Member 2.1.3 Clerk	
	2.2	System Features 2.2.1 Use Case Diagram 2.2.2 Domain Model: Class Diagram 2.2.3 State Diagram	
	2.3	Launch Phase	
	2.4	User Story Details	
		2.4.1 US001: User Story Register as Member User Story description of US001 Activity Diagram of US001 System Sequence Diagram of US001 2.4.2 US002: User Story View KADA Annual Report User Story description of US002 Activity Diagram of US002 System Sequence Diagram of US002 2.4.3 US003: User Story Apply Loan	

		User Story description of US003 Activity Diagram of US003 System Sequence Diagram of US003 2.4.4 US004: User Story View Loan Application Status User Story description of US004 Activity Diagram of US004 System Sequence Diagram of US004 2.4.5 US005: User Story View Individual Report User Story description of US005 Activity Diagram of US005 System Sequence Diagram of US005 2.4.6 US006: User Story Edit Profile User Story description of US006 Activity Diagram of US006 System Sequence Diagram of US006 2.4.7 US007: User Story Validate Application User Story description of US007 Activity Diagram of US007 System Sequence Diagram of US007 2.4.8 US008: User Story Update Application Status User Story description of US008 Activity Diagram of US008 System Sequence Diagram of US008 2.4.9 US009: User Story Update Dividend Details User Story description of US009 Activity Diagram of US009 System Sequence Diagram of US009	
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		2.4.10 US010: User Story View System Report User Story description of US010 Activity Diagram of US010 System Sequence Diagram of US010	
	2.5	Performance and Other Requirements (Non Functional)	
	2.6	Design Constraints	

1. Introduction

1.1 Purpose

This SRS document describes a single subsystem, for instance, document upload module and reporting module. The intended audience for the SRS is KADA employees.

1.2 Scope

The KADA System is a document management system (DMS) which able to solve problem faced by KADA currently. The main purpose of this system is to digitize and automate the flow of member registration, loan application and reporting as the current management system, paper-based document management system is less efficiency. Otherwise, transition to digitize system not only helps to improve the efficiency of workflows but also ensuring the data security for protecting user's information and improve user convenience. In another word, this system is a critical step toward transforming KADA into a modern, efficient, and technology-driven organization.

1.3 Definitions, Acronyms and Abbreviation

This subsection provides definitions for all terms, acronyms, and abbreviations used in this System Documentation (SD). The terms include key concepts central to our system's design and functionality, while the acronyms and abbreviations are listed with their full meanings to eliminate ambiguity and enhance understanding.

Terms

Agile	A software development methodology for iterative and incremental, focusing on collaboration and stakeholder feedback.
Persona	A fictional representation of user type, used to guide system design by capturing user requirements, needs and behaviors.
Use Case	A description of how a user interacts with the system to achieve a specific goal.
User Story	A brief description of a feature from the user's perspective.

Acronym

KADA	Lembaga Kemajuan Pertanian Kemubu (Kemubu Agricultural Development Authority)
SRS	System Requirements Specification: A document that captures the software's functional and non-functional requirements.
UML	Unified Modeling Language: A standardized modeling language used to visualize the design of a system.

Abbreviations

SD	System Documentation
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1.4 References

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3. TechAway, *PROJECT PROPOSAL AND PLANNING* (2024). Available: [https://github.com/ZHIMING04/KADA-Techaway/blob/main/SAD%20PROJECT%203%20\(1\).pdf](https://github.com/ZHIMING04/KADA-Techaway/blob/main/SAD%20PROJECT%203%20(1).pdf)
4. J. Berman, “Agile frameworks,” *Planview*, <https://www.planview.com/resources/guide/what-is-agile-program-management/agile-frameworks/> (accessed Dec. 5, 2024).
5. Department of Public Service Malaysia, *Malaysian Government Interoperability Framework (MyGIF), Version 1.0*, [Online]. Available: <https://www.kkr.gov.my/sites/default/files/jawatankuasa-ict/Standards%2C%20Policies%20and%20Guidelines%20-%20Malaysian%20Government%20Interoperability%20Framework%20%28MyGIF%29%20Version%201.0.pdf> [Accessed: Dec. 5, 2024].

1.5 Overview

The remainder of this SD includes the following:

- A description of Persona that specifies the characteristics and needs of intended user of the KADA eServe system in general
- System features with supporting diagrams such as the use case diagram, class diagram, and state diagram for better understanding of KADA eServe
- Product backlog outlines sprint plans and task assignments based on the use case diagram.
- Detailed description of the user story for each use case, stimulate various scenarios, detailing specific needs with supporting diagram such as sequence diagram and activity diagrams.
- Non-functional requirements that declare the characteristics or qualities of the system.
- Design constraints defined by the stakeholders.

This SD follows ISO/IEC/IEEE 29148 standard guidelines to ensure clarity, consistency, and usability.

2 Specific Requirements

This section outlines the software requirements in detail to guide the process of design, development, and testing. By describing all required inputs, outputs and functionality in specifics, it ensures that the system satisfies the needs of users and stakeholders.

2.1 Persona

This section introduces the target users of the system, focusing on their characteristics and needs. Understanding the personas helps ensure that the system is appropriate for its target audience.

2.1.1 KADA User

KADA users represent a diverse group from the general public, with different levels of technical ability and familiarity with technology and online platforms.

User Need

- A KADA user needs a way to apply to become a member in an easy and efficient manner.

User Stories

- As a KADA user, I want an accessible interface with clear instructions so that I can navigate the system without confusion.
- As a KADA user, I want a simple digital form to register to become a KADA member.
- As a KADA user, I want to receive an email regarding the update for my application status.
- As a KADA user, I want to see the KADA annual report to know about the company's annual financial and other information.

2.1.2 KADA Member

KADA member is an entity after KADA user register to become a member. These users can use the system for more features compared to KADA user. They can apply for loans, manage their savings and get an individual report about the loan and savings details. They have limited technical expertise with basic knowledge of using computers such as browsing websites or using simple applications. Before this, they used traditional paper-form mode for managing the member register and loan application process rather than using digitize system for this process.

User Need

- A KADA Member needs a way to submitting loan application to apply loan in more digitize and convenience way.
- A KADA Member needs a way to receive updates and notifications to stay informed about the application status.
- A KADA Member needs a way to view individual report to check for their personal savings and loan details.
- A KADA Member needs a way to view annual report to know about the company's annual financial and other information.

- A KADA Member needs a way to learn using the system to familiar with the system feature for using it smoothly.

User Stories

- As a KADA Member, I want to submit a loan application online so that I don't need to submit the form physically.
- As a KADA Member, I want to view and receive notifications about my loan status so that I am promptly informed of updates.
- As a KADA Member, I want an interface with adjustable text sizes and a simple layout so that I can use the system comfortably regardless of my technical skills.
- As a KADA Member, I want to view my savings balance and loan details so that I can keep track of my personal financial.

2.1.3 Clerk

Clerk is an entity that will verify updated application details, gather weekly applications for board of directors' review, updates website details such as dividend details, and compiles annual reports using collected data. Clerk is the one that has authority to edit and update the KADA eServe system. They possess higher technical expertise compared to KADA users and KADA members, primarily in using the KADA eServe application's own features and basic office tools for data handling, reporting, and system updates.

User Need

A clerk needs a way to digitize forms to minimize storage issues, reduce the risks of data loss, and improve efficiency in managing records.

User Stories

- As a clerk of KADA, I want to centralize digital platform to manage member registrations, loan applications, and savings so that I can reduce paperwork, minimize errors, and improve efficiency in serving employee KADA.
- As a clerk of KADA, I want to check and update the status of applications so that I can ensure accurate and timely communication with employee KADA.
- As a clerk of KADA, I want to generate reports on member registrations, loan applications, and savings so that I can provide insights for better organizational planning.
- As a clerk of KADA, I want the system to securely store all data so that the information is protected from loss or unauthorized access.
- As a clerk of KADA, I want a user-friendly interface for the system so that I can quickly adapt to the digital workflow without extensive training.

2.2 System Features

The KADA eServe system is a software system designed to operate on mobile and desktop devices, including both Android and iOS operating systems. The system provides KADA members with an efficient way to manage their financial activities, including loan applications, savings tracking, and report access, while simplifying the administrative processes for KADA staff, client. Additionally, the system utilizes a locally hosted database to enhance data retrieval speed, ensure security, and provide a seamless user experience tailored to KADA's operational needs.

2.2.1 Use Case Diagram

The system features are illustrated in Figure 1 below. The detailed description of each module and functions is tabulated in Table 1.

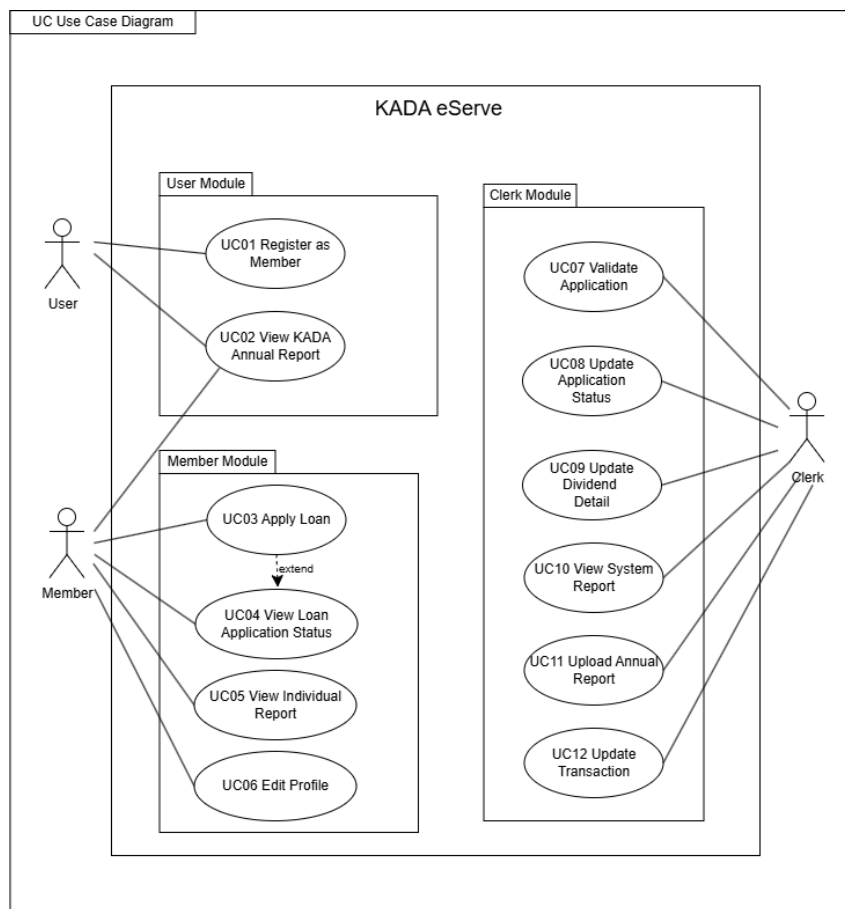


Figure 1.1: Use Case Diagram for KADA eServe

Table 1: Description of Module and Functions for KADA eServe

Use case	Function	Description
UC01	Register Member	KADA users can sign up as a member under KADA eServe system by filling up the online form.

UC02	View KADA Annual Report	KADA users and members would be able to view the KADA annual report.
UC03	Apply Loan	KADA member can apply for different types of loan by filling up the online form.
UC04	View Loan Application Status	KADA member can check their loan status to keep track on their loan application progress.
UC05	View Individual Report	KADA member would be able to view their individual report to have a clear sight on their membership , loan and financial details.
UC06	Edit Profile	KADA member would be able to edit their personal information on profile page.
UC07	Validate Application	Clerk would be able to check and validate the information of membership and loan application.
UC08	Update Application Status	Clerk would be able to update the status of the membership and loan application status in the system.
UC09	Update Dividend Details	Clerk would be able to update the dividend rate for updating the savings calculation.
UC10	View System Report	Clerk would be able to view the management reports on member registrations, loan applications, and savings for better organizational planning.
UC11	Upload Annual Report	Clerk would be able to upload the KADA's annual report for

Functional Requirement of Each Use Case

UC01 Register as member

REQ 1: The system shall provide a dedicated registration page accessible to visitors.

REQ 2: The system shall display registration form

REQ 3: The system shall validate all form inputs to ensure completeness and correctness.

REQ 4: The system shall store the registration details in the local database upon success validation.

REQ 5: The system shall generate a unique Member ID for each registered user.

REQ 6: The system shall notify user of successful submission with a confirmation message.

REQ 7: The system shall display error messages for invalid input or duplicate records.

UC02 View KADA annual report

REQ1: The system shall provide a dedicated section on the website for users to access the annual report.

REQ2: The system shall display a list of available annual reports, including metadata such as year and title.

REQ3: The system shall display a user-friendly error message, If the requested report is not available.

REQ4: The system shall allow the administrator to upload the new annual report through the security management panel.

UC03 Apply loan

REQ-1: The system should provide an interface for members to submit loan applications.

REQ-2: The system should notify the clerk of pending loan applications for verification and allow the clerk to approve or reject loan applications.

REQ-3: The system shall calculate the loan amount, interest rate, and monthly payment schedule upon approval.

REQ-4: The system shall notify the member of the loan application status, whether approved, pending or rejected.

REQ-5: The system shall display the calculated details to the member.

REQ-6: The system shall store the loan repayment details in database for tracking.

UC04 View loan application status

UC05 View individual report

REQ 1: The system shall provide a “Report” button in the navigation bar.

REQ 2: The system shall display the “Annual Report” list and “Individual Report” button.

REQ 3: The system shall display the details of the individual report page (e.g., Savings, Total Loan, Dividends, Loan Repayment Dates, Membership Start Date, Loan Application Success Dates, and the statuses of Membership and Loan Applications).

REQ 4: The system shall display a user-friendly error message if the requested visitor is a non-KADA member.

UC06 Edit Profile

REQ 1: The system shall allow members to access their profile details from the dashboard.

REQ 2: The system shall provide an interface for users to edit their personal information.

REQ 3: The system shall validate the updated fields to ensure data accuracy and adherence to predefined format.

REQ 4: The system shall provide a “Save Changes” button to commit the updated profile details to the database.

REQ 5: The system shall display a confirmation message upon successfully saving the profile changes

REQ 6: The system shall allow users to cancel the editing process and revert to the original profile details without saving any changes.

REQ 7: The system shall ensure that users can only edit their own profiles and prevent unauthorized access to other users’ profiles.

REQ 8: The system shall display an error message if the database update fails, notifying the user to retry later.

UC07 Validate application

REQ 1: The system shall provide a "Validate Applications" button in the navigation bar.

REQ 2: The system shall display a list of pending membership and loan applications when the clerk clicks the "Validate Applications" button.

REQ 3: The system shall allow the clerk to review the application details to ensure the accuracy and completeness of information.

REQ 4: The system shall allow the clerk to mark the application as:

- **Validated** if all criteria are met.
- **Rejected** if missing information or documents are identified, with a comment field for specifying the reason of rejection

REQ 5: The system shall allow the clerk to resume the validation process once the applicant resubmits their complete application.

REQ 6: The system shall notify the clerk if a technical error occurs during the validation process.

REQ 7: The system shall store a record of the validation activity for auditing and tracking purposes.

UC08 Update application status

REQ 1: The system shall provide a "Update Application Status" button in the navigation bar.

REQ 2: The system shall display a list of applications with their current statuses (e.g., Pending, Validated, Rejected).

REQ 3: The system shall require the clerk to provide comments when updating the application status to "Rejected."

REQ 4: The system shall prompt the clerk to enter comments if they attempt to select "Rejected" without providing comments.

REQ 5: The system shall update the application status in the database upon submission

REQ 6: The system shall notify the applicant about the status update and the next steps if the status is "Rejected" via an e-mail

REQ 7: The system shall store a record of the status update in the system for auditing and tracking purposes.

UC09 Update dividend details

REQ 1: The system shall provide an "Update dividend details" button in the navigation bar.

REQ 2: The system shall provide a column for updating the dividend rate.

REQ 3: The system shall verify the entered dividend rate to avoid an illogical value being updated, such as a negative value or an extremely large value.

REQ 4: The system shall prompt the clerk to decide whether the entered rate is correct before it is successfully updated to the whole system.

REQ 5: The system shall notify the clerk if the update is unsuccessful, or the update rate is illogical.

REQ 6: The system shall store a record of the latest dividend details in the system for calculating savings for members.

REQ 7: The system shall update the dividend details and savings of members in their own individual report.

UC10 View System Report

REQ 1: The system shall provide a summary of member registrations, including the success and failure rates, along with demographic details.

REQ 2: The system shall offer insights into the number and status of loan applications, approval or rejection rates, and related financial summaries.

REQ 3: The system shall display data on member savings, including the total savings, dividend details, and *growth trends*.

REQ 4: The system shall analyze and present savings trends, showing year-over-year in member savings behavior.

REQ 5: The system shall allow data export in formats like Excel or PDF for external analysis or reporting purposes.

2.2.2 Domain Model: Class Diagram

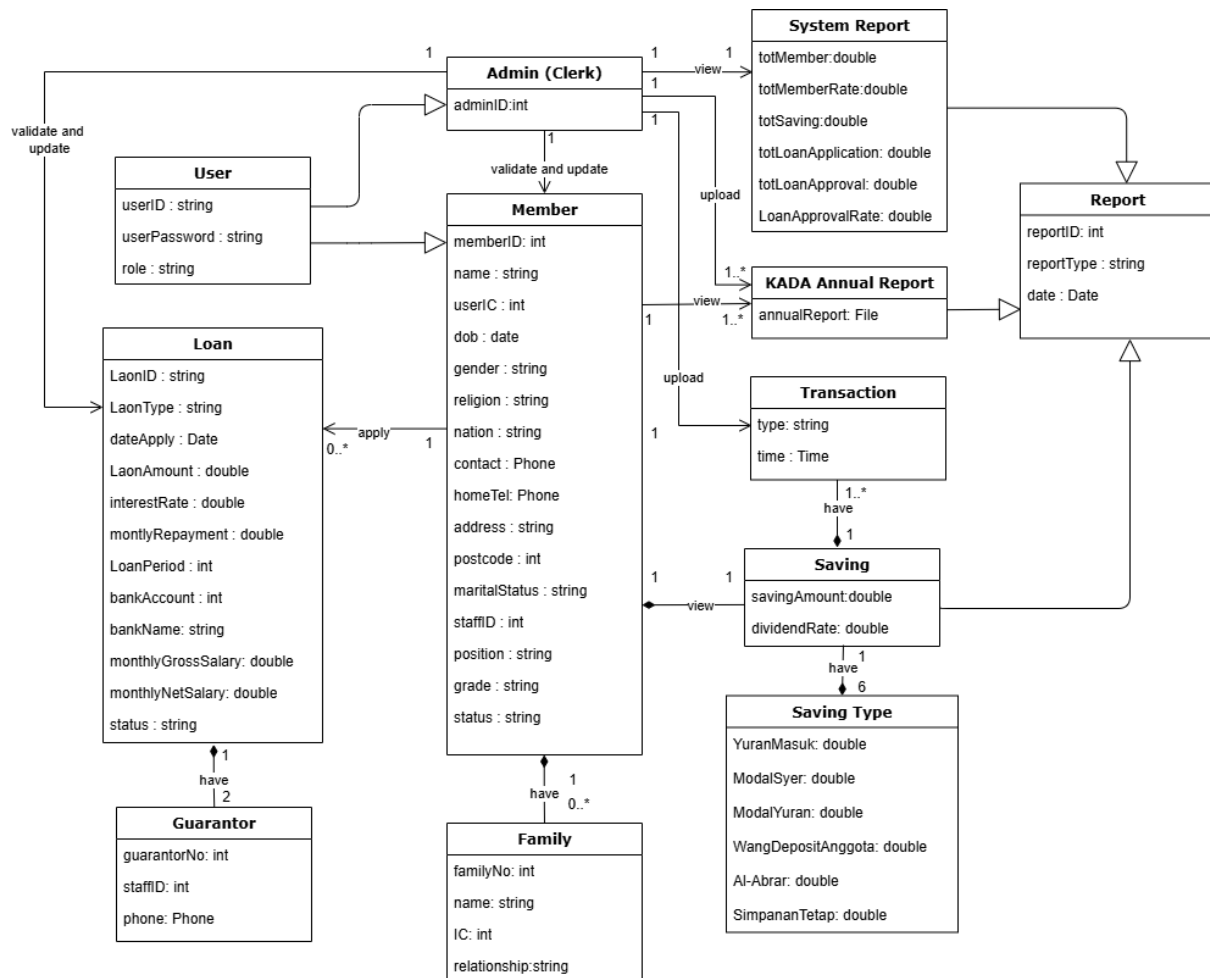


Figure 1.2: Domain Model for KADA eServe

User Class

Represent all users that use the system. This class's contents will serve as login credentials, fundamental user data, and a breakdown of login responsibilities.

- **userID:** A unique identifier for the user to login to the system.
- **userPassword:** Password to login to the system.
- **role:** Represent the two user roles that can be accessed in the system: administrator and member.

Member Class

Represent a person registered in the system as a member. As a user applies for a member, an object of the member is created.

- **memberID:** A unique identifier for the member.
- **name:** Name of the member.
- **userIC:** The identification number of member (MyKad).
- **dob:** Date of birth of member.

- gender: Gender of member. This attribute accepts predefined value such as Male (Lelaki) and Female (Perempuan).
- religion: Religion of member. This attribute accepts predefined values such as Islam, Buddhism (Buddha), Hinduism (Hindu), Christian (Kristian) and Others (Lain-lain).
- race: Race of member. This attribute accepts predefined values such as Malay (Melayu), Chinese (Cina), Indian (India) and Others (Lain-lain).
- nation: Nation of member. This attribute accepts predefined values such as Malaysian or Non Malaysian.
- homeTel: Home number of member.
- address: Address of member.
- postcode: Postcode where member live.
- maritalStatus: Marital status of member.
- staffID: KADA staff id of member.
- position: Member's position under KADA company.
- positionGrade: Position's grade of the member.

The Member Class is inherited from the User Class. A member whose status is accepted would be able to apply for a loan.

Admin Class

Represent administrative users (clerks) who manage applications and update their statuses based on decisions made by the Board of Directors (BOD).

- adminID: A unique identifier for the admin.

The Admin class is inherited from the User Class. An admin would be able to validate and update the status of the Application class which included member application and loan application.

Guarantor Class

Represent a guarantor for the loan application.

- guarantorNo: A unique identifier for the guarantor.
- staffID: KADA staff ID of the guarantor.
- phone: Phone number of the guarantor.

The Guarantor class is a composition of the Loan class, where there must be two guarantors for each application.

Family Class

Represent the family member of user

- familyNo: A unique identifier for the family member.
- name: Name of the family member.
- IC: Identity card (MyKad) number of the family member.
- relationship: Relationship of the family member with the member.

The Family class is composition of the Member class. The relationship is one to many relationships, where a member can have zero or more family members.

Application Class

Represent types of applications submitted, such as member applications and loan applications

- applicationID: A unique identifier for application.
- applicationType: Represent the type of application which are member application and loan applications.
- date: Date of latest updated application status.
- status: The status of application.

Loan Class

Represent loan applications submitted by members, including detailed financial information and repayment details.

- LoanID: A unique identifier for loan.
- LoanType: Represent the type of loan that member able to apply.
- dateApply: Date when application submitted.
- LoanAmount: The amount of loan applied by the member in RM.
- interestRate: The interest rate applicable to the loan.
- monthlyRepayment: The monthly repayment amount of the applied loan in RM.
- LoanPeriod: The repayment period of the applied loan.
- bankAccount: The bank account of the member.
- bankName: The bank associated with the bank account number provided.
- monthlyGrossSalary: The monthly gross salary of the member in RM.
- monthlyNetSalary: The monthly net salary of the member in RM.
- memberID: served as foreign key to

A member fulfilling all the criteria would be able to apply for loans.

Report Class

Serves as a general representation of all reports within the system, including financial, operational, and performance-related documents.

- reportID: A unique identifier for the generated report.
- reportType: Represent the type of report.
- date: Date when report is generated.

Saving Class

Represent the member's savings details

- savingAmount: The total amount of member's savings in RM.
- dividendRate: The dividend rate of savings.

The Saving Class is inherited from Report class. The members can view their own saving details.

KADA Annual Report Class

Represent the yearly financial summary of KADA, capturing profit, loss, and key highlights.

- annualReport: The KADA annual report that will be uploaded in pdf file form.

The KADA Annual Report Class is inherited from Report class. The user and member can view one or many KADA annual report.

System Report Class

Represent the monthly and yearly summary of

- totMember: The total number of members.
- totSaving: The total number of members' saving amount.
- totLoanApplication: The total number of loan applications.
- totLoanApproval: The total number of approval loans.
- activeAccount: The total number of members that are active
- inActiveAccount: The total number of members that are inactive.

The System Report Class is inherited from Report class. The admin can view the system report.

2.2.3 State Diagram

The state diagram show the states and transitions of an object throughout the system. The Application class would require a state diagram that shows the transition of application status.

Figure 1.3 and figure 1.4 shows the state diagrams for the Application class that involves two types of application which are member application and loan application. Once the user submits the application, the application will be in Processing state. After the meeting with Board of Directors, the result of the application will be updated to either Approve state or Rejected state. As the application is approved, the applicant will be a member of KADA.

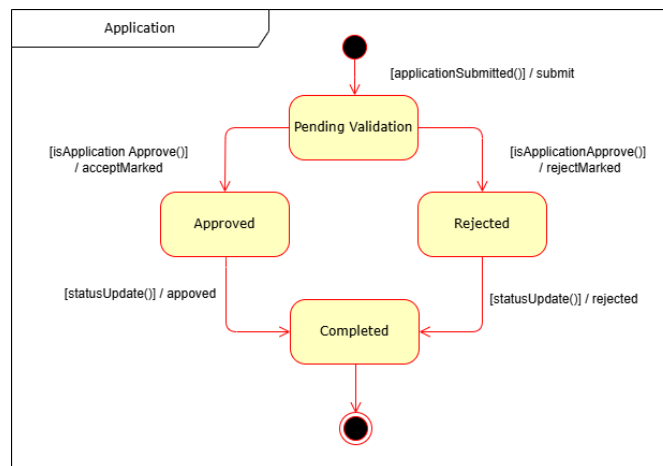


Figure 1.3 State Machine Diagram for member and loan application

2.3 Launch Phase

Sprint	Team members assigned
Sprint #1	
Sprint #2	
Sprint #n	

2.4 User Story Details

This section outlines the user's goals and expectations for the system, focusing on the required actions. It defines the user story's scope, flow of events, sequence diagrams, and activity diagram to ensure the system effectively meets user's needs.

2.4.1 US001: User Story Register as Member

Table 2.4.1: User Story Description for Register as Member

User story: Register as Member
ID: US001
User Story Description As a visitor to the KADA's website, I want to register myself as a KADA member, So that I can access additional functions that are exclusively for members, such as applying for loans.
Flow of events: 1. Visitor accesses the registration page. 2. Visitor fills in the registration form with required details. 3. Visitor submits the registration form. 4. The system validates the form details. 5. Upon successful validation, the system stores the member data in the database and displays a success message. 6. System notify the status of application to user.
Alternative flow : 1. If the user submits invalid or incomplete information, the system will highlight the errors and request corrections. 2. If duplicate records are detected, the system will notify the user and prevent duplicate registrations.
Acceptance Criteria Postcondition • The user becomes a registered member and is assigned a unique number ID. Precondition • The visitor has access to the KADA eServe registration page other conditions • The database must be connected to the local database to store registration data.
Exception flow: • If the database connection fails, system displays an error message and asks user to try again later.

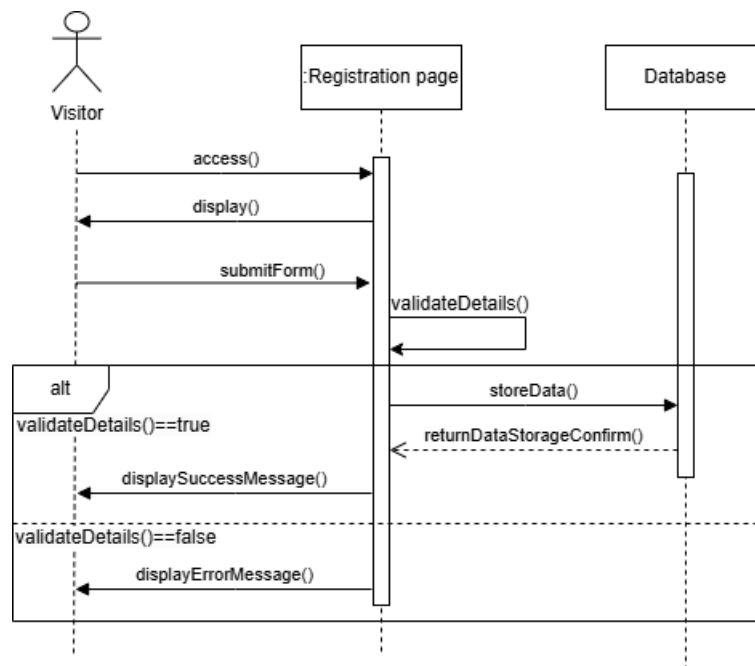


Figure 2.4.1.1: Sequence Diagram for Register as Member

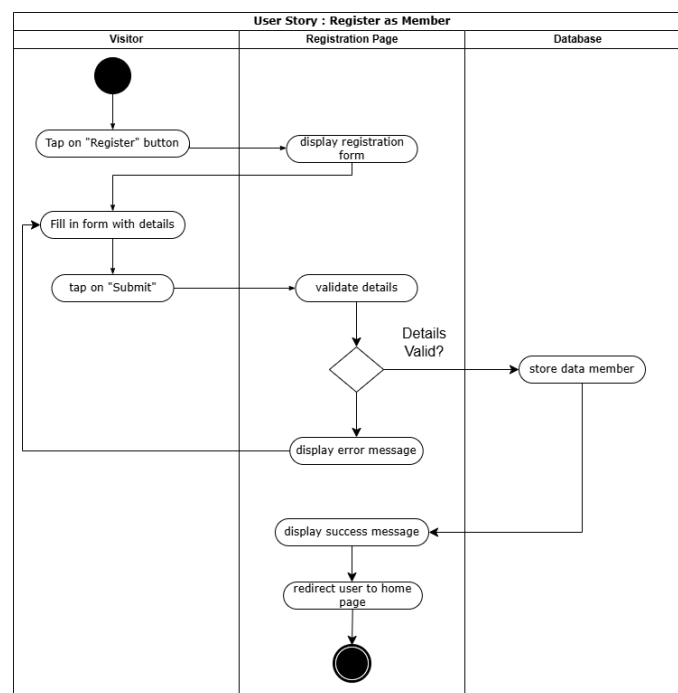


Figure 2.4.1.2: Activity Diagram for Register as Member

2.4.2 US002: User Story View KADA annual report

Table 2.4.2: User Story Description for View KADA annual report

User story: View KADA annual report
ID: US002
User Story Description
As a visitor/member to the KADA's website, I want to view the KADA's annual report, So that I can understand their financial status, achievements, and future goals.
Flow of events: 1. <u>User Accesses the Website</u>: opens KADA's website via a web browser. 2. <u>User Navigates to the Reports Section</u>: clicks on menu item "Reports". 3. User Selects a Specific Report: clicks on the "Annual Reports". 4. System Displays the Report: The system opens the report in a new page
Acceptance Criteria <i>Postcondition:</i> 1. Users can navigate to the annual reports section. 2. The page content can be properly display across devices. 3. Users can easily understand the purpose and content from its label or title. <i>Precondition:</i> 1. The website has a dedicated section for annual reports, accessible from the navbar. 2. The annual report is uploaded to the system and saved in a readable format. 3. The report tagged with appropriate metadata.
Exception flow: 1. Display friendly error message if file is currently not available.

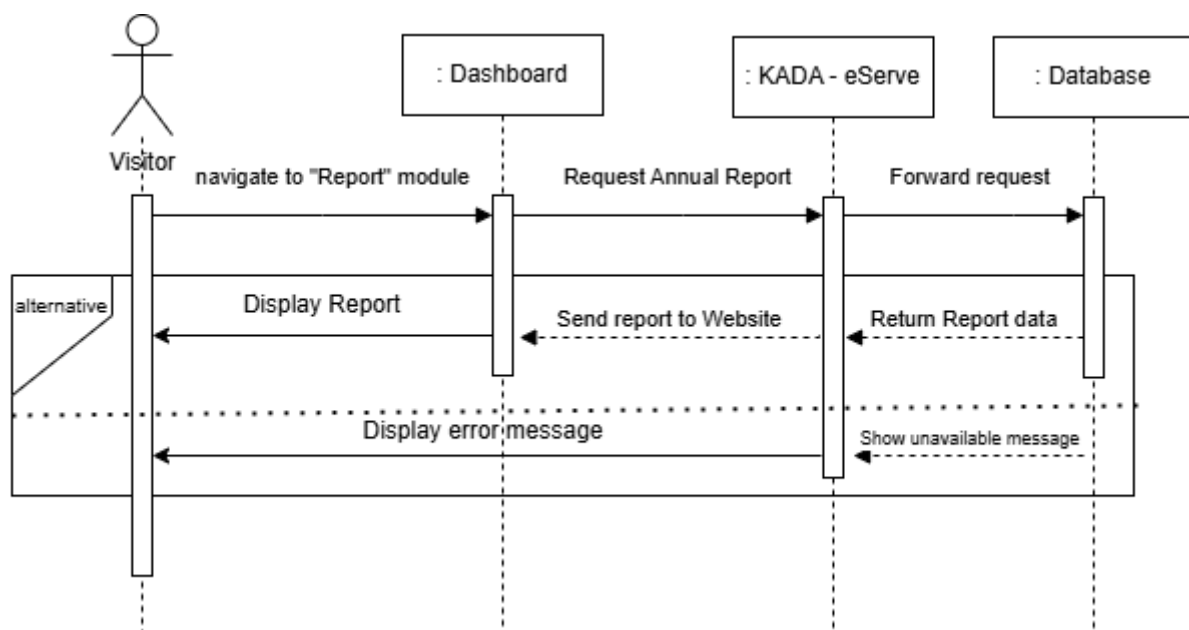


Figure 2.4.2.1: Sequence Diagram for View Annual Report

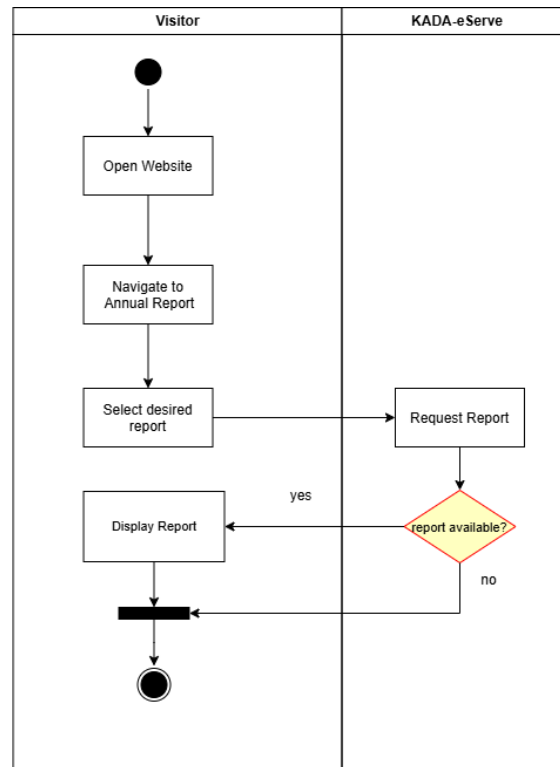


Figure 2.4.2.2: Activity Diagram for View Annual Report

2.4.3 US003: User Story Apply loan

Table 2.3: User Story Description for Apply loan

User story: Apply Loan
ID: US003
User Story Description As a member to the KADA website I want to access the loan application So that I can get loan for my business
Flow of events: 1. The member logs into the KADA eServe platform. 2. The system displays the loan application form in the Member Module. 3. The member selects the loan type and enters necessary details, such as the loan amount, purpose, and repayment period. 4. The system validates the information entered and checks the member's eligibility based on predefined criteria 5. If all criteria are met, the system processes the loan application and generates an acknowledgment message. 6. The system notifies the member of the application status.
Alternative flow: ● Flow 1 (Invalid Information): The system notifies the member of the error and asks them to update any incomplete or incorrect loan information they have entered. ● Flow 2 (Ineligible Member): If the member fails to meet the loan eligibility criteria, the system displays a notification indicating the reason for rejection and suggests corrective actions (if applicable).
Acceptance Criteria Postcondition ● The loan application is successfully submitted and stored in the system for further processing. ● The member receives a notification regarding the application status. Precondition ● The KADA system account must be active for the member. ● To apply for a loan, the member must fulfill the minimal eligibility requirements. Other conditions ● The system must perform validation checks in real-time to ensure data accuracy. ● Loan limits and repayment terms must align with KADA's financial policies.
Exception flow: ● System Error: The member is asked to try again later if the system finds a problem during submission, and the application is not saved. ● Eligibility Check Failed:

The system notifies the member and logs the cause for rejection if the member does not meet the eligibility requirements.

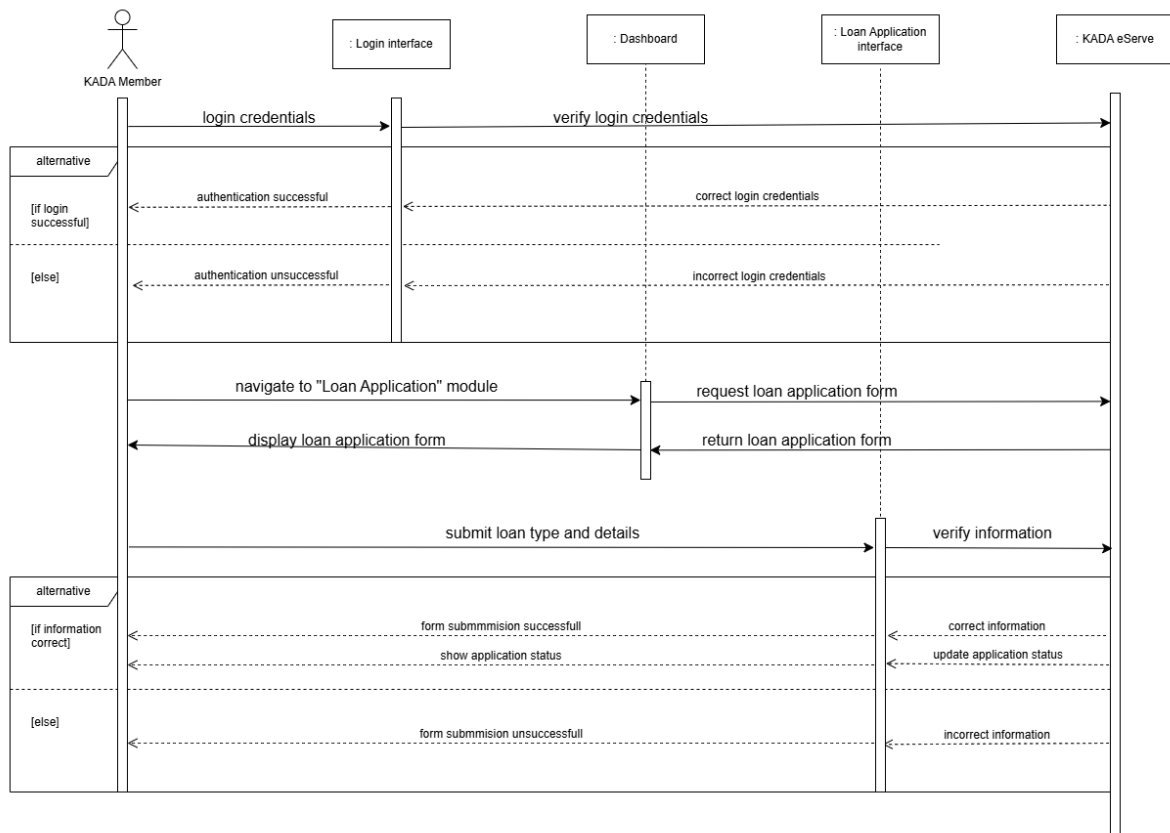


Figure 2.3.1: Sequence Diagram for Apply loan

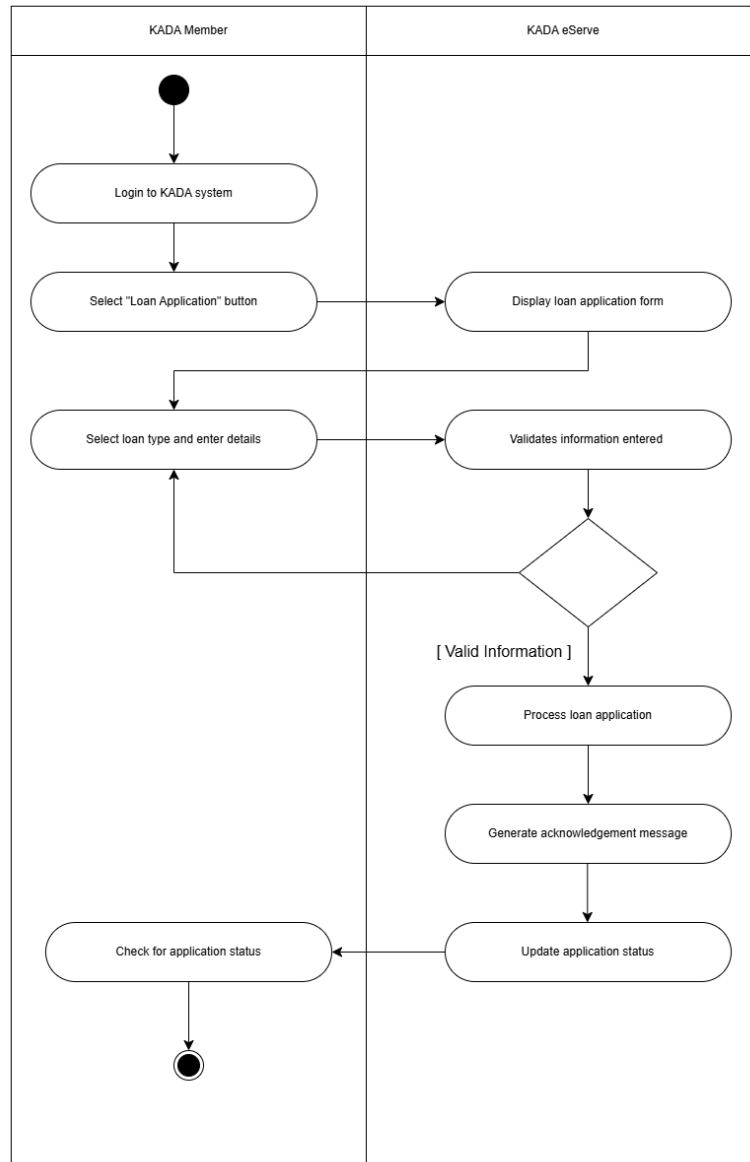


Figure 2.3.2: Activity Diagram for Apply Loan

2.4.4 View Loan Application Status

Table 2.4: User Story Description for View Loan Application Status

User story: View Loan Application Status
ID: US008
User Story Description <i>As a</i> member of KADA, <i>I want to</i> view my loan application status <i>so that</i> I can get to know the current loan application status.
Flow of events: 1. The user logs into the system and navigates to the “Status” section. 2. The system retrieves the current member loan status from database. 3. The system displays the current status timeline at status page.
<u>Acceptance Criteria</u> Postcondition • The application status is updated in the system. • An email is sent to the applicant with details about the updated status and next steps. • A record of the update is stored in the system. Precondition • The clerk must have valid login credentials and access to the "Update Application Status" module. • The application must exist in the system.
Exception flow: E1: Notification Unsuccessfully Sent • The system fails to send an email to the applicant about the status update (e.g., due to an invalid email address). • The system logs the failure and retries the notification process.

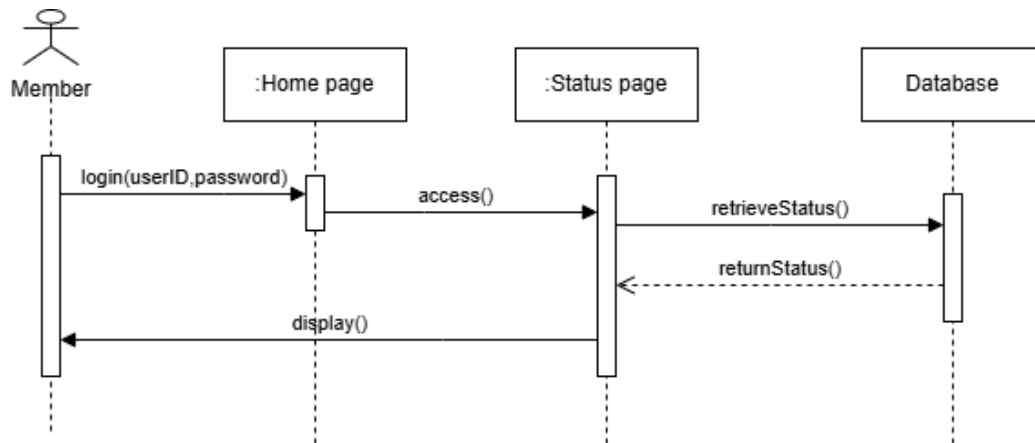


Figure 2.4.1: Sequence Diagram for View Loan Application Status

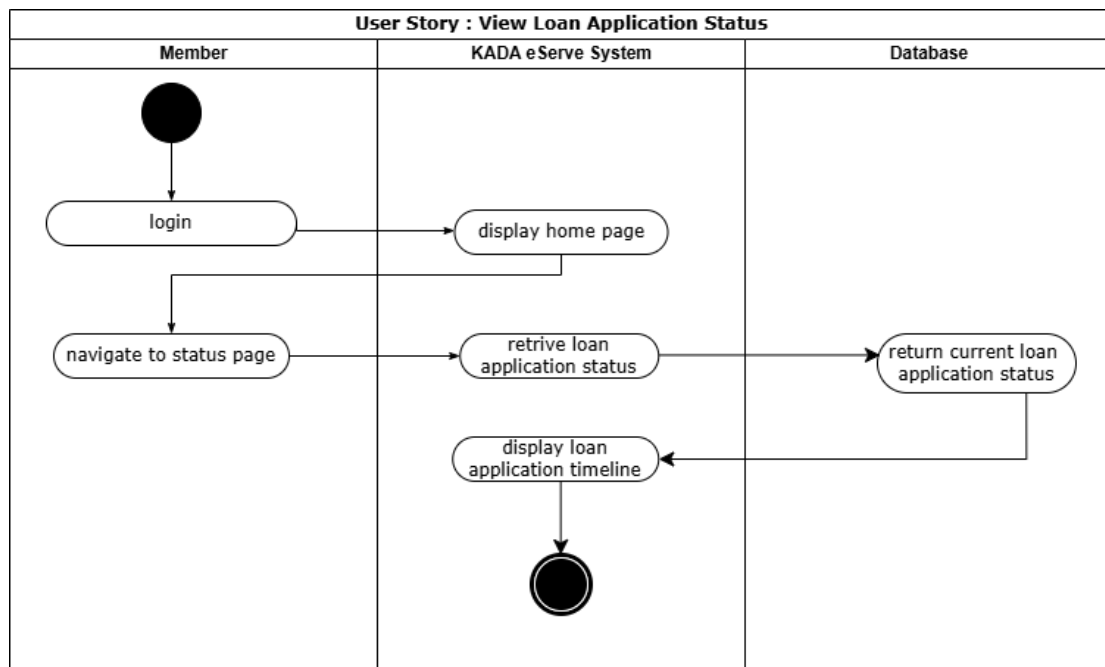


Figure 2.4.2: Activity Diagram for View Loan Application Status

2.4.5 US005 User Story View Individual Report

Table 2.5: User Story Description for View Individual Report

User story: View Individual Report
ID: US005
User Story Description As a member to the KADA website I want to view my individual report So that I can understand a summary of my membership and financial details.
Flow of events: 1. Member logs into the KADA eServe website. 2. Member navigates to the report page via the navigation bar or a report icon on the home page. 3. Member selects the “Individual Report”. 4. System redirects the member to their individual report page and displays all their membership and financial details.
Alternative flow: ● Flow 1 (Loan Application Pending/Unsuccessful): If the member has not yet applied for a loan, their loan application is unsuccessful or has not been approved, the system will not display the full individual report instead of the approved part. ● Flow 2 (Non-Member Access): If the user is not a registered member of KADA, the system denies access to the individual report and displays an access denied message. They can only access KADA’s annual report, which is available to the public.
Acceptance Criteria Postcondition ● The individual report is displayed with the correct data to the members, including all details about their membership and savings, even if they have not applied for a loan. ● If the member has not applied for a loan, the loan-related column will not be shown in the report. ● The report is formatted properly and displayed across devices for easy reading. Precondition ● The user must be a registered member in KADA eServe system to access the individual report. ● The member must meet specific eligibility criteria such as active membership. ● The member’s loan application must be approved for loan details to appear in the report.
Exception flow: ● The system will notify the member with an error message to advise them to retry later.

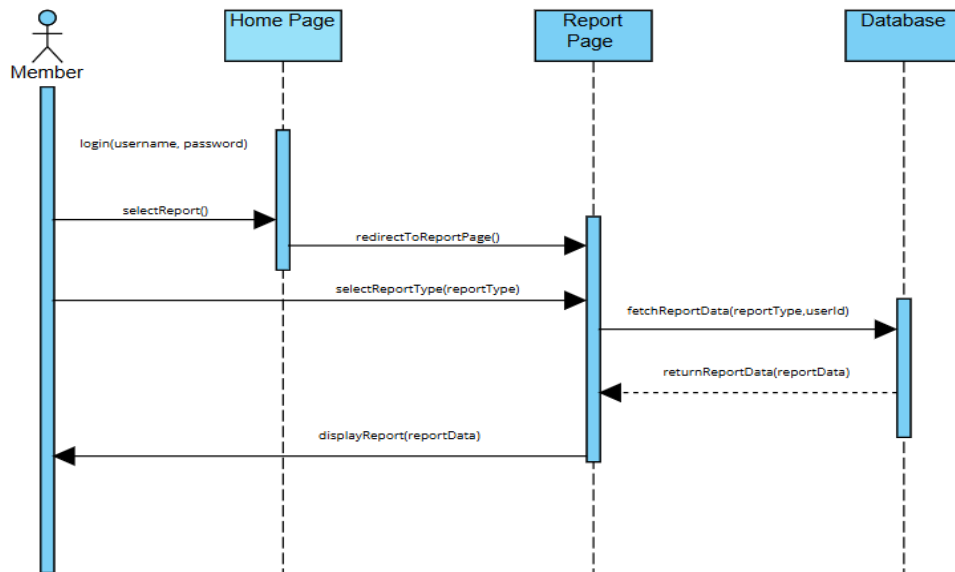


Figure 2.5.1: Sequence Diagram for View Individual Report

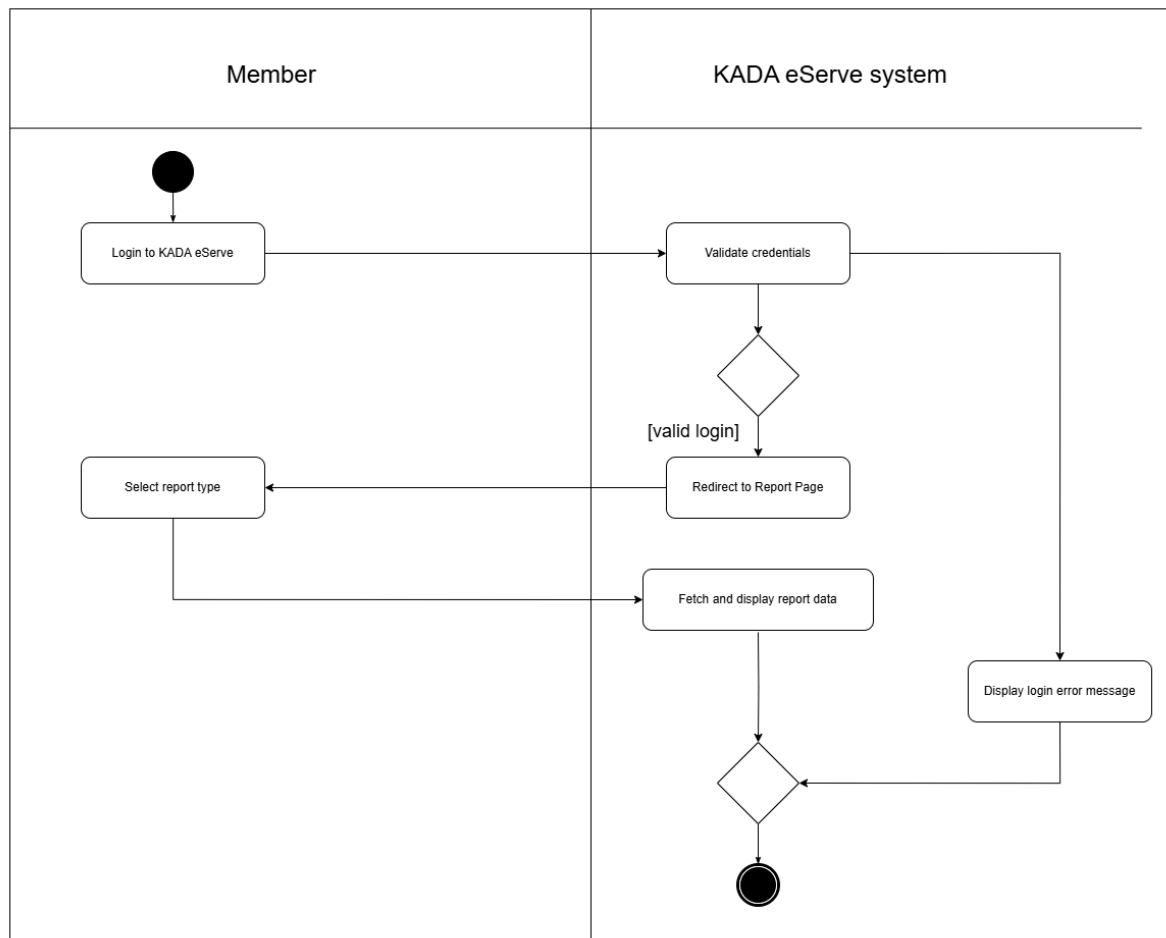


Figure 2.5.2: Activity Diagram for View Individual Report

2.4.6 US006 User Story Edit Profile

Table 2.6: User Story Description for Edit Profile

User story: Edit Profile
ID: US006
User Story Description As a member to the KADA website I want to edit y profile So that I can update my personal information in the system
Flow of events: 1. The user logs into the system and navigates to the “Profile” section for editing profile. 2. The system retrieves the user’s current profile details from the database and displays them. 3. The user modifies the desired fields 4. The user clicks the “Save Changes” button to submit the updated profile. 5. The system validates the inputs, ensuring all fields meet required formats. 6. The system updates the user’s profile in the database and displays a confirmation message.
Alternative flow: ● Flow 1 (Invalid Input): If the user enters invalid data, the system displays an error message and highlights the problematic fields. The user corrects the input and resubmits the changes. ● Flow 2 (Cancel Changes): If the user chooses to cancel the update process, the system reverts the displayed information to the original profile details.
Acceptance Criteria Postcondition 1. The user’s profile is updated in the database with the new details. 2. An audit log is created, recording the changes and the timestamp. Precondition 1. The user must be logged into the system. 2. The system must have the user’s profile information stored in the database.
Exception flow: ● Database Update Failure: If the database update fails due to a technical issue, the system notifies the user with an error message and advises retrying later. ● Unauthorized Access: If the system detects unauthorized attempt to edit another user’s profile, it deicnd access and logs the incident.

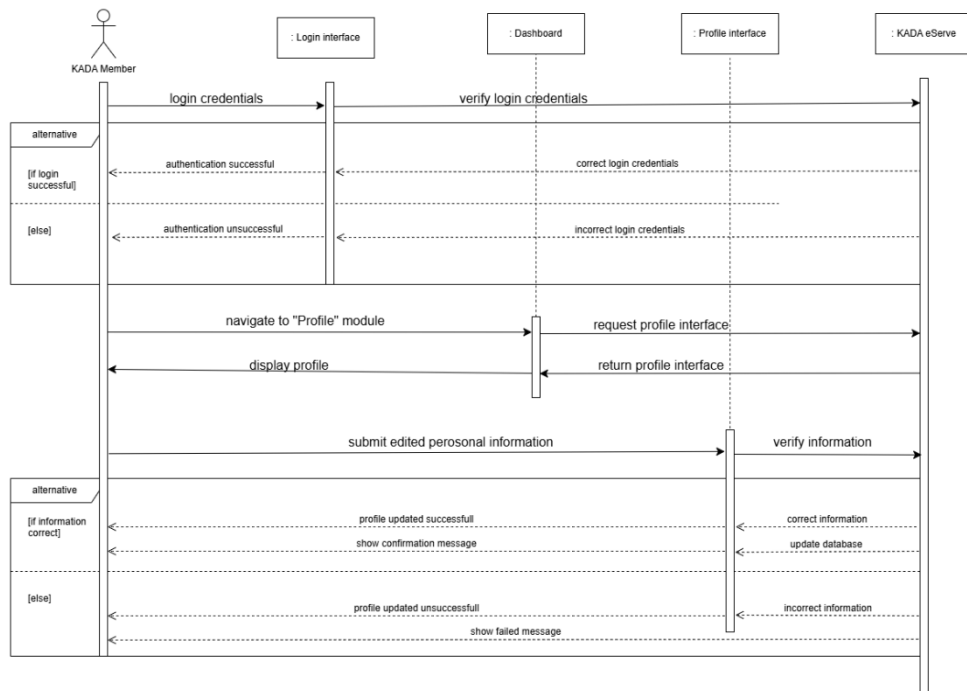


Figure 2.6.1: Sequence Diagram for Edit Profile

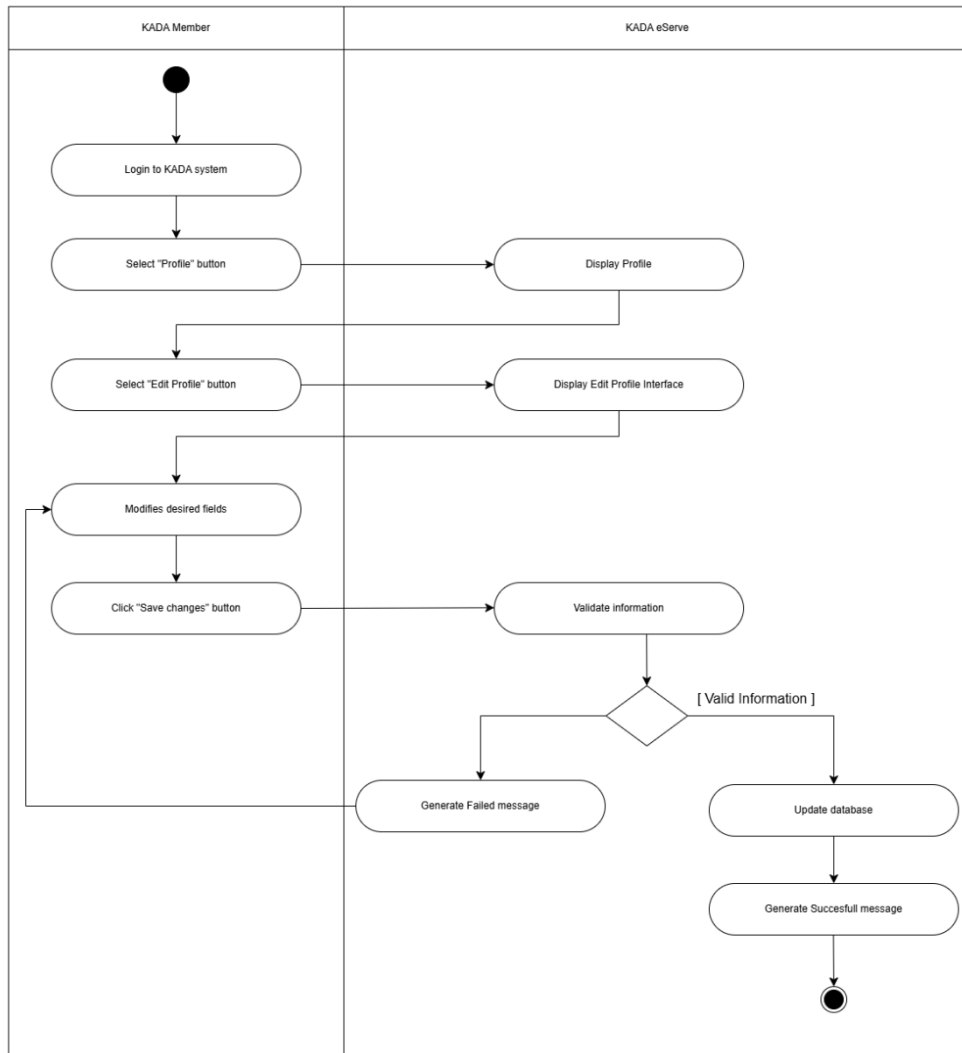


Figure 2.6.2: Activity Diagram for Edit Profile

2.4.7 US007 User Story Validate Application

Table 2.7: User Story Description for Validate Application

User story: Validate Application
ID: US007
User Story Description <i>As a</i> clerk of KADA, <i>I want to</i> be able to validate the information in membership and loan applications <i>so that</i> I can ensure the accuracy and compliance of customer submissions.
Flow of events: 1. The clerk accesses the KADA eServe system through the login interface using their username and password. 2. The system authenticates the clerk and directs to the dashboard 3. The clerk clicks on the “Validate Applications” button from the navigation bar 4. The system displays a list of pending membership and loan applications 5. The clerk selects a specific application from the list. 6. The system retrieves and displays the complete application details, including uploaded documents. 7. The clerk carefully reviews the application to ensure: • The provided personal and financial information is complete and accurate. • All required documents have been submitted 8. If the application meets all criteria, the clerk marks it as “Validated” in the system. 9.If issues are found, the clerk marks it as “Rejected” and provides comments specifying the required changes or missing details.
Alternative flow: A1: Missing documents in application 1.The clerk identifies that one or more required documents are missing. 2.The clerk marks the application as “Requires Corrections” in KADA eServe 3.The system notifies the applicant to upload the missing documents. 4.The clerk resumes validation once the applicant resubmits the complete application.

Acceptance Criteria

Postconditions

1. The application status is set to one of the following:
 - a. *Validated*:
If the application meets all accuracy and compliance requirements.
 - b. *Rejected*:
If errors or missing information are identified.
2. Notifications are sent to the applicant with the validation outcome and next steps (if corrections are required).
3. A record of the validation activity is stored in the system for auditing and tracking purposes.

Preconditions

1. The clerk must have valid login credentials and be logged into the KADA eServe system.
2. The application to be validated must be available in the "Validate Applications" module.
3. All required documents for the application must be uploaded by the applicant.
4. The system must be functional and accessible to retrieve and display application details.

Exception flow:

E1: Technical Issue with KADA eServe

1. The clerk encounters a system error while validating an application.
2. The clerk reports the issue to the technical support team.
3. The clerk pauses validation until the issue is resolved.
4. The system provides an update when the error is resolved, allowing the clerk to proceed with the validation process.

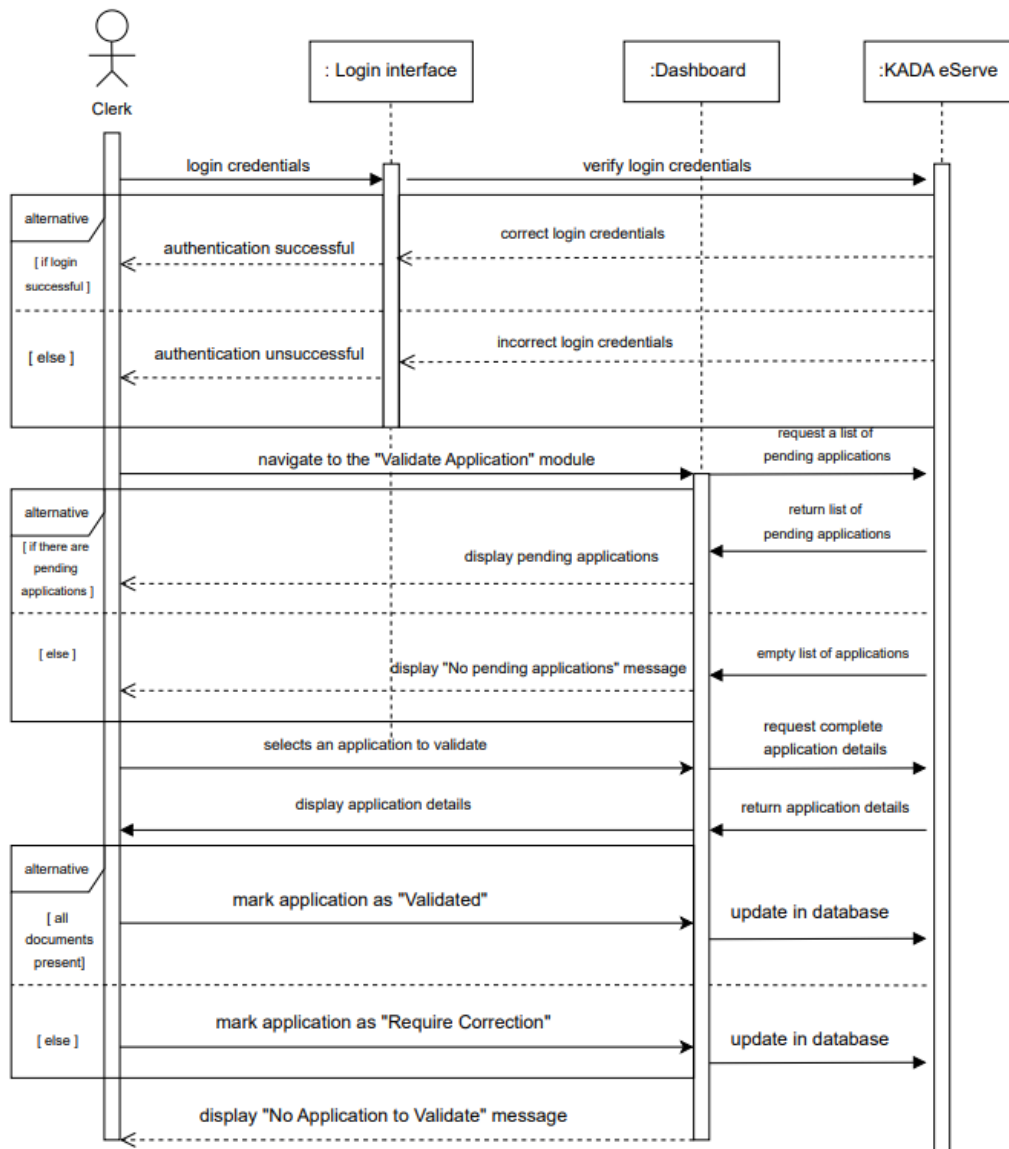


Figure 2.7.1: Sequence Diagram for Validate Application

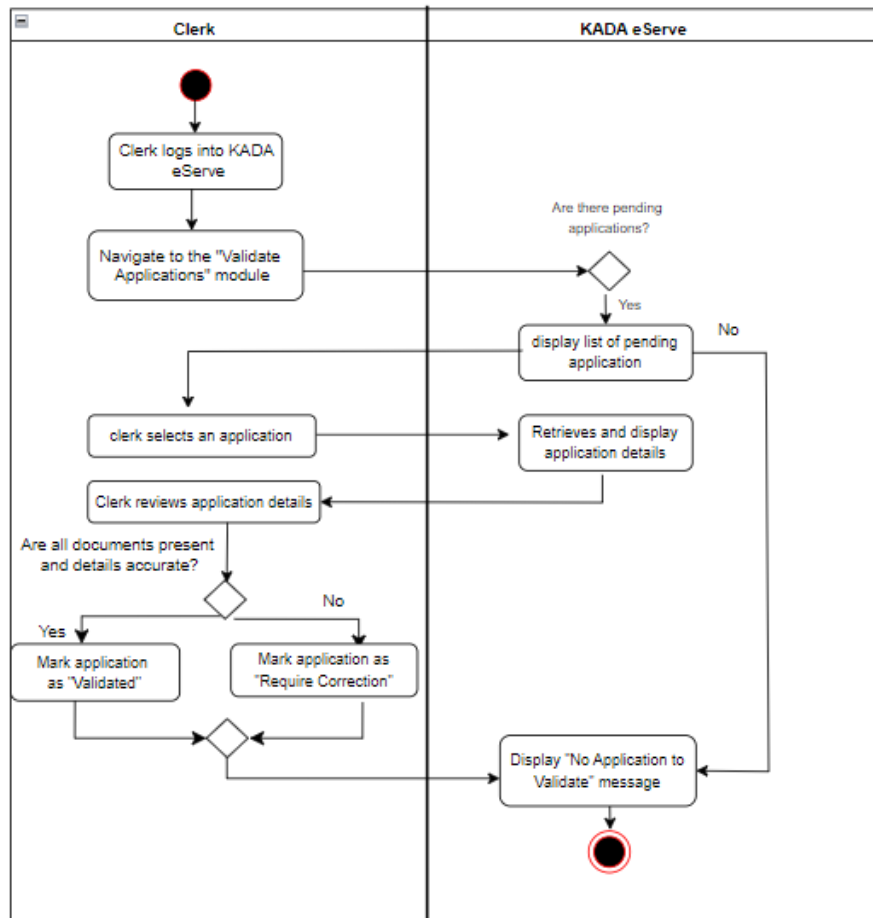


Figure 2.7.2: Activity Diagram for Validate Application

2.4.8 US008: User Story Update Application Status

Table 2.8: User Story Description for Update Application Status

User story: Update Application Status
ID: US008
User Story Description <i>As a</i> clerk of KADA, <i>I want to</i> be able to update the application status of the membership and loan applications <i>so that</i> I can reflect the progress of each application and ensure applicants are notified of the application results
Flow of events: 1. The clerk accesses the KADA eServe system using their username and password. 2. The system authenticates the clerk and navigates to the dashboard 3. The clerk clicks on the “Update Application Status” button from the navigation bar. 4. The system displays a list of application with their statuses (eg: Pending, Validated, Rejected) 5. The clerk selects a specific application to update its status. 6. The clerk chooses a new status from the available options (e.g., Validated, Requires Correction, Rejected). 7. If the clerk selects "Rejected," they must provide comments specifying the reason for the status change 8. The system updates the status in the database and stores the action as part of the application's history. 9. The system notifies the applicant about the status update via a notification in the applicant portal.
<u>Alternative flow :</u> A1: Missing Comments for Rejected Status 1. The clerk attempts to update the application status to "Rejected" without providing comments. 2. The system prompts the clerk to enter comments before updating the status.
<u>Acceptance Criteria</u> Postcondition • The application status is updated in the system. • An email is sent to the applicant with details about the updated status and next steps. • A record of the update is stored in the system.

Precondition

- The clerk must have valid login credentials and access to the "Update Application Status" module.
- The application must exist in the system.

Exception flow:

E1: Technical Issue with KADA eServe

- The clerk encounters a system error while updating a status
- The clerk reports the issue to the technical support team using the "Support Request" feature in "KADA eServe".
- The clerk pauses validation until the issue is resolved.
- The system provides an update when the error is resolved, allowing the clerk to continue the validation process.

E2: Notification Unsuccessfully Sent

- The system fails to send an email to the applicant about the status update (e.g., due to an invalid email address).
- The system logs the failure and retries the notification process.

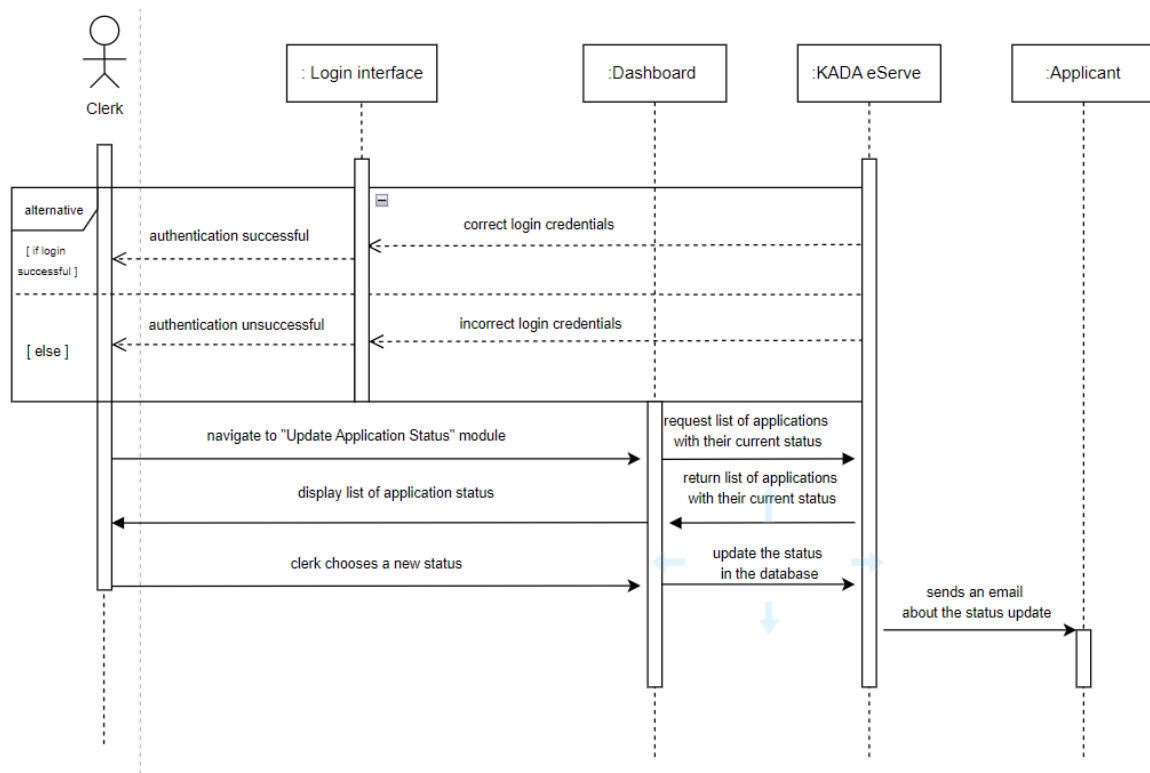


Figure 2.8.1: Sequence Diagram for Update Application Status

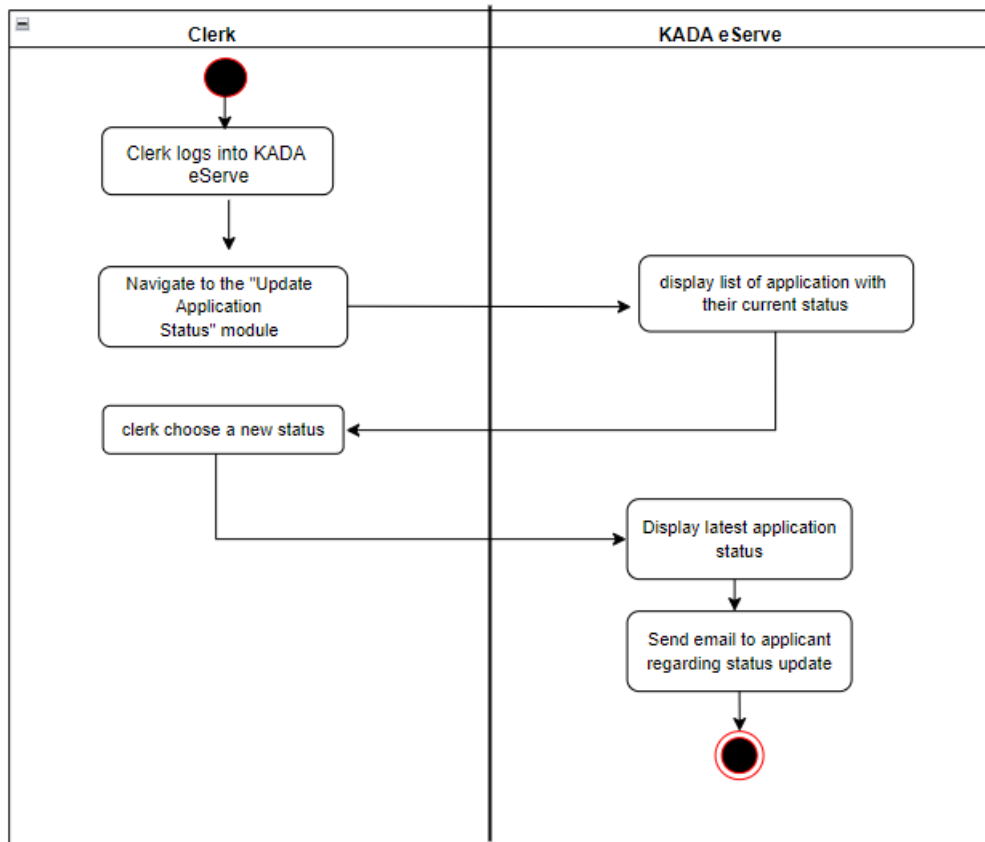


Figure 2.8.2: Activity Diagram for Update Application Status

2.4.9 US009 User Story Update Dividend Details

Table 2.9: User Story Description for Update Dividend Details

User story: Update Dividend Details
ID: US009
User Story Description As a clerk of the KADA I want to update the dividend details that have been calculated So that the system can display the updated dividend rate on the individual report and KADA's annual report and calculate the savings for members accordingly.
Flow of events: 1. Clerk logs into the admin page of KADA eServe system. 2. The clerk navigates to the dividend update section of the system. 3. The clerk enters the new dividend rate and updates the system. 4. The system updates the dividend rate and recalculates any affected member savings. 5. The system saves the updated dividend details in the system. 6. The system displays updated dividend details in both the individual reports and the annual report, ensuring accurate savings calculations for every member.
Alternative flow: ● Flow 1 (Invalid Dividend Rate): If the clerk enters an invalid or unreasonable dividend rate such as negative values or extremely high rate, the system will display an error message to remind the clerk. The update will not be saved until a valid rate is entered. ● Flow 2 (System Error): If there is a system error while updating the dividend rate, the system will prompt out a message to notify of an unsuccessful update. The clerk will update after the system issue is solved. ● Flow 3 (Non-Member Access): If the user is not a registered member of KADA or does not have permissions, they will not be able to access the dividend update feature.
Acceptance Criteria Postcondition ● The individual report is updated with a new dividend rate and applied to member savings. ● The updated dividend details are displayed in both the individual report and annual report. ● The report is formatted correctly and displayed the updated dividend details in the specified sections. ● The system ensures the new dividend rate is applied to member calculations properly, and any changes are accurately reflected across the system. Precondition

- The clerk must have permission to log into the edit feature of website to update dividend details.
- The clerk must be logged into the KADA eServe system with valid credentials.
- The dividend rate must be updated correctly based on company policy, and the system should perform necessary validation checks before saving the rate.

Other condition

- The updated dividend details must be stored in the system and made available for reporting purposes.
- If the dividend details update successfully, the system should notify the clerk of the successful update message.

Exception flow:

- The system will notify the clerk if the system encounters an issue while updating the dividend details.
- If the dividend rate entered is invalid, the system should display an error message.
- If the user is not an authorized clerk, they will not have permission to log into the system's edit feature to update the dividend details.

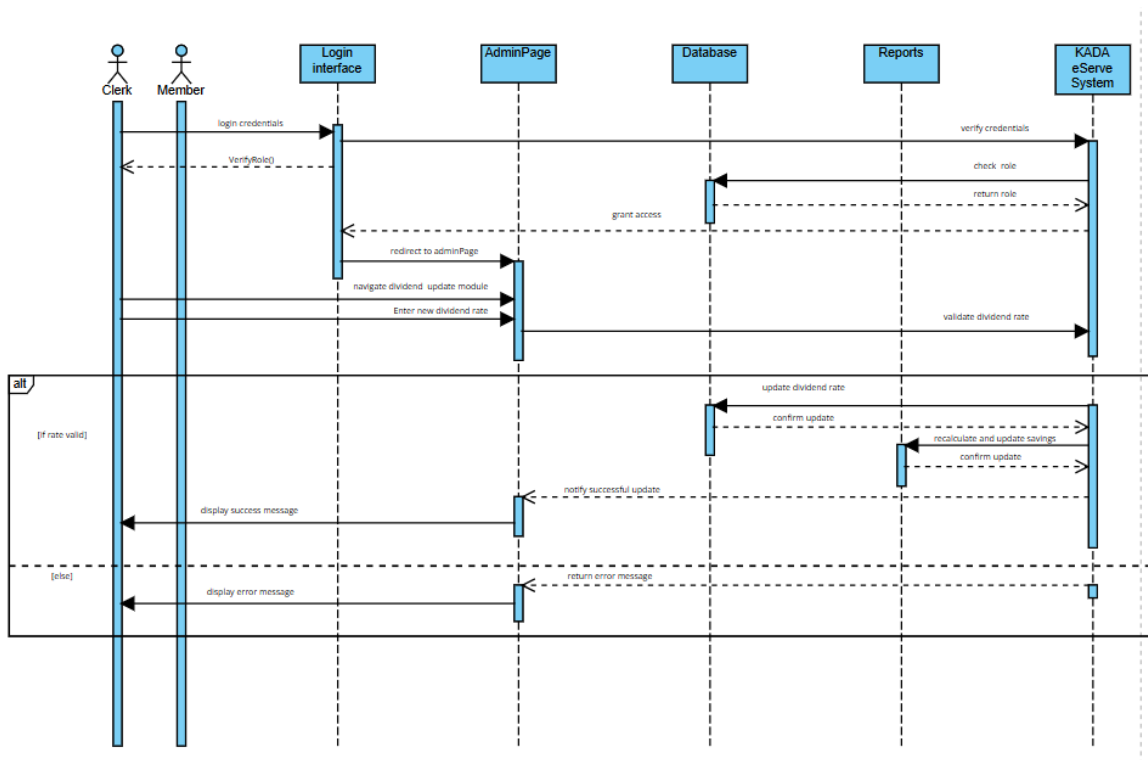


Figure 2.9.1: Sequence Diagram for Update Dividend Details

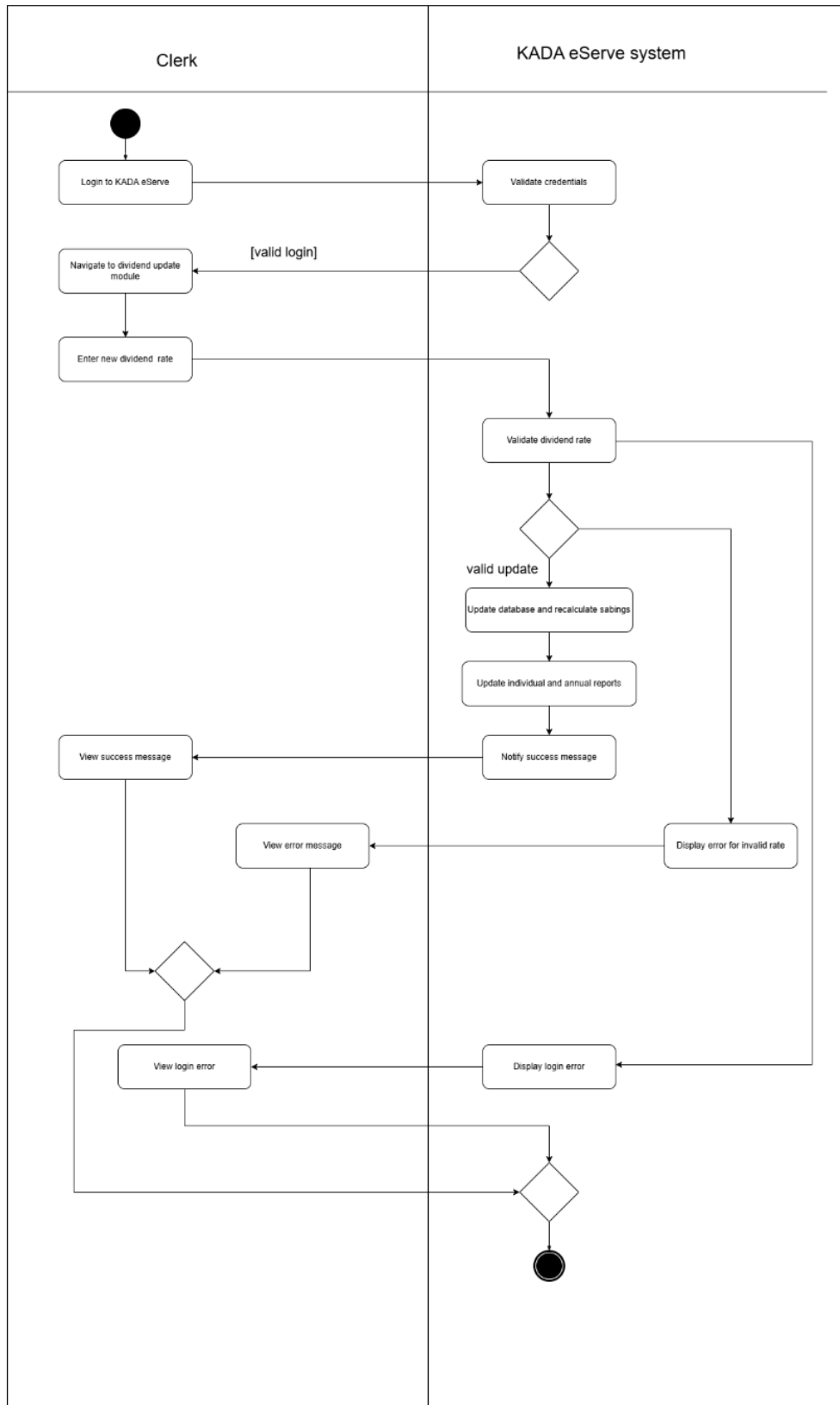
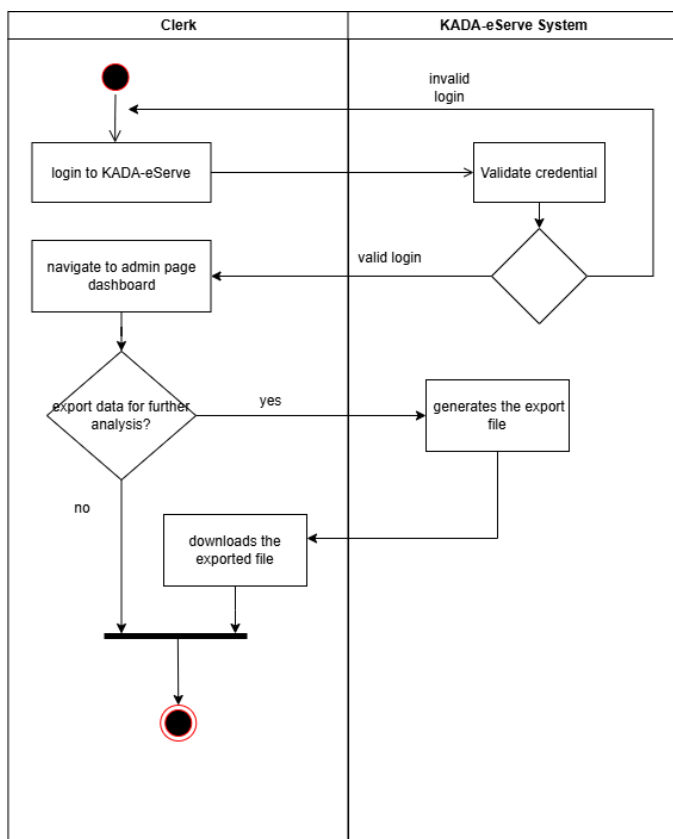
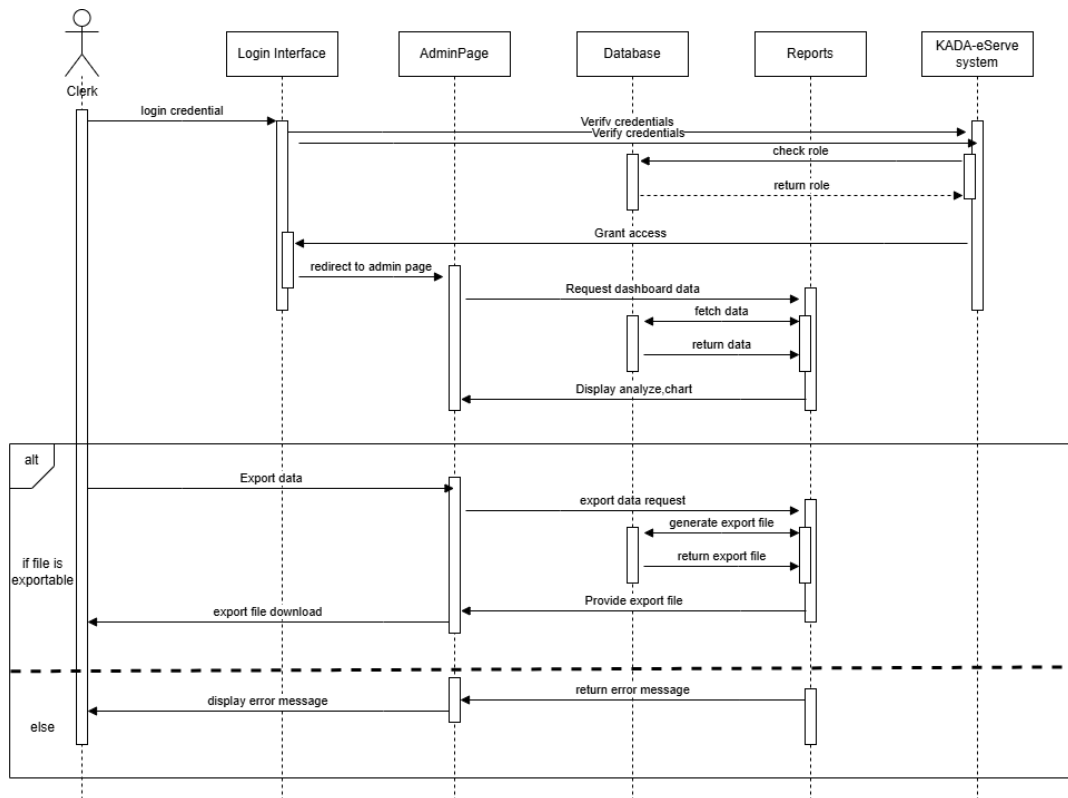


Figure 2.9.1: Sequence Diagram for Update Dividend Details

2.4.10 US010 View System Report

Table 2.10: View System Report

User story: View System Report
ID: US010
User Story Description As a clerk of the KADA I want to easily understand the general contents through some simple analysis So that I can better take appropriate measures against the corresponding results.
Flow of events: 1. Clerk logs into the admin page of KADA eServe system. 2. Clerk will access to the dashboard by default. 3. Clerk view member registration summary. 4. Clerk review loan application insight. 5. Clerk analyst member saving data. 6. Clerk export data for further analysis.
Alternative flow: • Flow 1 (Export Failure): Network or server issues during data export, the system displays an error message: "Export failed. Please try again later." and the clerk retries or contacts support.
Acceptance Criteria Postcondition • The clerk successfully views the system reports or exports them for external analysis. Precondition • The system contains the latest data on member registration, loan applications and member savings. Other condition • The clerk successfully views the system reports or exports them for external analysis.
Exception flow: • The clerk successfully views the system reports or exports them for external analysis.



2.5 Performance and Other Requirements

The KADA system is designed to ensure efficient performance and optimal user experience. The following performance requirements define its capability to respond to user requests and manage data effectively:

Response Time: The system shall retrieve and display document search results within a maximum of 3 seconds for typical search queries and within 5 seconds for complex queries.

Throughput: The system must handle up to 200 concurrent requests per minute without a significant decline in performance.

Capacity: The system can handle 1,000 concurrent users without performance degradation and store up to 500,000 records in the database.

Availability: The system must maintain 99.5% uptime, ensuring it is always operational and accessible except during scheduled maintenance.

Software Quality Attributes

Maintainability: The system should be designed to be easy to maintain, enabling developers and administrators to troubleshoot, update, and extend the system. This includes clear, well-documented code, modular architecture, and the use of standard design patterns.

Reliability: The system must be highly reliable to ensure that members have uninterrupted access to their information and services. This includes the robustness to handle expected and unexpected situations without causing system failure. Periodic monitoring and automatic failure recovery mechanisms should be implemented to ensure availability.

Efficiency: The system must deliver a high level of performance, ensuring fast response times and resource optimization, especially during peak loads. The system should process large amounts of data, such as membership registrations and loan applications, without affecting performance.

Reusability: Systems should be designed as components and modules that can be reused across different parts of the system or in future projects. By following principles like modularity and abstraction, developers can minimize duplication of effort and ensure that system components can be reused as needed.

Usability: Availability is critical to ensuring that all user classes, whether administrators or members, can interact effectively with the system. The user interface should be intuitive, easy to navigate, and accessible to all users. It should minimize the steps required to complete tasks such as signing up, applying for a loan, and viewing savings. The system should also provide useful feedback to users to guide them through the process.

2.6 Design Constraints

Environmental constraints: The system must be designed to operate within a temperature range of 10°C to 30°C, typical for office or server room environments. It should be able to function properly in environments with humidity levels ranging from 30% to 70%. The system should also be capable of withstanding environments with minimal dust and particulate matter to prevent hardware damage. Besides, the system must ensure stable performance with a minimum internet speed of 5Mbps and be capable of operating in environments with intermittent network connectivity.

Compatibility constraints: Since Windows 10 and later versions are the main platform for development and deployment, the system must run without a hitch on time. Additionally, the system will support modern web browsers, including Google Chrome (v100 and above), Mozilla Firefox (v95 and above), and Microsoft Edge (v90 and above). It will function well on systems that have 256 GB of SSD storage, an Intel i5 processor or similar, and at least 8GB of RAM. For data management, the system is fully compatible with MySQL (v8.0) as its primary database. Furthermore, it aligns with the Laravel Framework (v10.x) and its associated PHP libraries to ensure stable and efficient development and deployment process. These compatibility constraints offer a strong basis for dependable and efficient system operation.

Security constraints: All sensitive information, including personal and financial data must be encrypted using HTTPS protocols for transmission. The system will enforce secure authentication processes, requiring users to log in with a username and password, supplemented by OTP-based two-factor authentication during registration and password recovery. Role-based access control (RBAC) will ensure that users can only access features relevant to their roles, such as Members and Clerks. To safeguard sessions. The system will automatically log out users after 15 minutes of inactivity, and all critical actions such as logins and data modifications, will be logged for monitoring. The system must implement regular updates to address vulnerabilities and maintain compliance with security standards, including Personal Data Protection Act (PDPA).

Performance Constraints: Under a load of 1,000 concurrent users, the system's response time for any user action, including file upload, search, or form submission, should not exceed 2 seconds. The system should be able to support up to 1,000 concurrent users without performance degradation.

Hardware Constraints: The system must be designed to run on a minimum of 8GB of RAM. The system must have enough memory and storage to support MySQL to handle database operations effectively. The system must run on a server with at least a quad-core processor (e.g., 3.0 GHz or higher) to handle concurrent user requests and backend processes. For scalability in the event that data volume increases, loan application documents and user data must be stored on an SSD with a capacity of about 100GB. The system must include redundant servers to ensure continuous ability and minimise downtime during hardware failure. To guard against cyberattacks and unauthorized access, the system needs hardware firewalls to guarantee the security of all KADA eServe users' data.